UF UNIVERSITY of FLORIDA
Herbert Wertheim College of Engineering

CIS4XXX

M2: Design Prototype (Application)

Overview

As a team, you will complete and deliver a ***design prototype*** demonstrating the feasibility and core features of your project. The features demonstrated will encompass a ***vertical slice*** – i.e., an iteration of your project that has all core features integrated in at least one dimension, though it need not be polished. The prototype should not involve implementations of all aspects of every feature; instead, the design prototype is intended to provide a basis for architecture and practical testing (including usability and user experience). The goal of the design prototype deliverable is to identify exactly how the project will be structured and to provide a foundation for the second phase of the project.

Specification

The deliverable should meet the following requirements.

User Experience

The user feedback system, including UI and sensory response elements, should be in place by this milestone. All elements of user control should be present.

Interface

All user interface elements should be visible and usable. Options should be intuitive and consistent. Additionally, the currently selected button(s), menu option(s), or other dialog element(s) should be highlighted. General status changes must be visible in the UI.

Navigation

All user mechanics should be functional. Controls should be thoroughly tested. Bugs should be minimal, if present, and must not detract significantly from the overall experience. Controls should not require significant time for acclimation; they should be predictable and easy to use. Features should be discoverable and well-explained.

User Perception

Application responsive and use must be intuitive to users. Teams should have solicited user testing from non-team members to ensure general usability. Users should report an enjoyable experience. The prototype should help users attain stated goals with minimal frustration.

Responsiveness

Interactions with the user interface must be indicated by the sensory experience (visually and/or audibly). Changes in application state should be indicated on or near the relevant interface elements. Additionally, completion or failure of tasks must be clearly indicated to the user.

Build Quality

This build should endeavor to avoid major bugs. All content, including art and code systems, should be present.

Robustness

By this milestone, crashes should not occur within the context of regular (unexceptional) use. There should also be no major or noticeable glitches within regular use.

Consistency

Except where explicitly otherwise by design, the application should act predictably, i.e., for the same input and use case, it should yield the same result. When and if the application behaves unpredictably, it must be for a clear and compelling reason.

Aesthetic Rigor

No major cosmetic issues should be present. Any aesthetic issues that are present should not cause the application to be unusable. All assets (e.g., images and sounds) should be present and functional within the context of the project.

Vertical Features

All major architectural elements should be complete with at least one demonstrative implementation.

Front-End

For each major use-case, at least one variant must be implemented within the UI of the application. This must connect to the *persistent state*.

Persistent State

For each major use case, there must be a demonstration of use of a data store in the application. This must connect to the *front-end* and the *back-end*.

Back-End

For each major use case, at least one variant of the data processing step must be implemented. This must connect to the *persistent state*.

Submissions

Your submission will be a tgz (gzipped tar) archive including the following:

- Source code and assets for submission
- README file

Source & Assets

The submission should include all source code, including PIP package build scripts (e.g., setup.py) as well as run scripts as necessary. The project should be runnable using an executable command without parameters and should be prepared by build scripts. For server applications, a command to initialize the server state for first run may be included. Submissions should also include all pre-built assets necessary for application function (e.g., pre-built database collections necessary for initial function and/or visual/audio assets).

README File

A README file, either in plaintext or markdown (UTF-8) should be included. The README should have, at a minimum, the following information:

- A description of the project
- Package name for PIP installation (desktop) or one-line build command (mobile)
- Link to the repository for the project (e.g., GitHub) [Note: if private, you must add instructors]
- For mobile projects, a link to the application package (APK)
- Run executable command and, if application, initialization command